

Sayali Deshmukh

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 4+ years of experience architecting and deploying end-to-end machine learning systems, large language models (LLMs), and real-time AI services across enterprise and consumer platforms. Adept at building production-grade solutions using PyTorch, TensorFlow, Hugging Face Transformers, and modern MLOps frameworks across cloud (AWS, GCP, Azure), distributed GPU environments, and edge devices. Specialized in transformer-based architectures (LLaMA 2/3, GPT, CLIP), recommendation systems (GEM, Andromeda), NLP (NER, sentiment analysis, document parsing), and responsible AI practices (SHAP, LIME, GDPR/KYC compliance). Proven track record in model optimization (quantization, ONNX, TorchScript), CI/CD automation, and deploying scalable AI solutions in fast-paced Agile/Scrum teams.

WORK EXPERIENCE

AI/ML Engineer | Snowflake, CA, USA

Nov 2023 to Current

- Contributed to development and deployment of ML models (classification, NLP, recommender systems) in Snowpark ML, improving in-database inference efficiency by 32% and reducing reliance on external data pipelines.
- Developed feature engineering workflows in Snowflake Data Cloud using SQL, Python, and Snowpark APIs, enabling the processing of 50+ TB of structured/unstructured data daily for ML use cases.
- Assisted in building retrieval-augmented generation (RAG) pipelines with Cortex AI embeddings and vector search, helping reduce hallucination rates by 27% in enterprise-facing AI applications.
- Implemented components of ETL → training → serving pipelines with Airflow and dbt, improving reproducibility and reducing pipeline latency by 40%.
- Integrated Hugging Face transformers into Snowflake ML workflows for custom LLM fine-tuning, supporting compliance with data governance requirements.
- Helped develop real-time inference services using PyTorch, TensorFlow, and Snowpark ML deployed on Kubernetes + Docker, achieving sub-100ms response times in production.
- Supported experiment tracking and model governance with MLflow and Snowflake dashboards, improving model deployment velocity by 25% while maintaining compliance with internal standards.
- Built Streamlit applications inside Snowflake to deliver interactive AI dashboards, empowering 300+ business users to explore ML insights without external tools.
- Contributed to MLOps workflows using Terraform, Argo CD, and Prometheus-Grafana monitoring, ensuring 99.99% uptime of ML-driven services.
- Collaborated with data engineers to ingest and unify multi-cloud data sources (AWS S3, Azure Data Lake, GCP BigQuery) into Snowflake, lowering integration costs by 35%.
- Helped enhance LLM-powered summarization and chatbot features via Cortex AI and vector re-ranking, improving customer support resolution times by 22%.
- Worked with product managers, data scientists, and clients to productize ML solutions using Snowflake's Native App Framework, enabling 5+ enterprise AI applications to scale globally.

AI/ML Engineer | IBM, India

Jan 2020– Mar 2022

- Developed and deployed scalable ML models for classification, forecasting, and anomaly detection using scikit-learn, XGBoost, and LightGBM, improving operational risk scoring accuracy by 31% across financial services and ERP clients (SAP, NetSuite).
- Designed enterprise-grade NLP systems using spaCy, BERT, NLTK, and Hugging Face Transformers for named entity recognition (NER), text classification, and document parsing to process unstructured reports, contracts, and user feedback.
- Built multilingual sentiment analysis models integrated into customer service platforms, enhancing automation and customer satisfaction tracking across English, Hindi, and Marathi.
- Engineered ETL pipelines using Apache Spark, PySpark, Presto, and Hive, integrated with DVC and Airflow for full ML lifecycle tracking, leading to a 45% improvement in time-to-deploy.
- Deployed ML services using Docker, TorchServe, and Triton Inference Server, managed on Kubernetes clusters hosted across AWS SageMaker, Azure ML Studio, and GCP Vertex AI, with real-time inference latency <80ms.
- Applied quantization (INT8 and 4-bit) and TorchScript optimization to accelerate deep learning models on cloud and ARM edge devices, improving performance by 40% on mid-tier GPU clusters.
- Implemented MLOps with MLflow, GitHub Actions, Jenkins, and CI/CD pipelines, streamlining version control, model tracking, and shadow/canary deployments into staging and prod.
- Monitored model performance and data drift using Prometheus, Grafana, and custom dashboards via Power BI and Tableau, triggering auto-retraining workflows as needed.
- Embedded Responsible AI practices including SHAP, LIME, and Fairlearn, and collaborated with legal/compliance teams to align systems with GDPR, RBI, KYC, and AML regulatory frameworks.
- Collaborated across Agile/Scrum teams including Product Managers, DevOps, QA, and Data Engineering to deliver end-to-end ML-powered features; actively participated in code reviews and sprint demos using Git, Bitbucket, and Jira.
- Led internal workshops on reproducibility, MLOps tooling, and containerization environments, upskilling 30+ junior engineers in modern AI deployment best practices.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Bash, Java, C, JavaScript, HTML5, CSS3

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, Reinforcement Learning (RL, RLHF), LLaMA 2/3, GPT, CLIP, Emu, EmuEdit, EmuVideo, GEM, Andromeda

Natural Language Processing (NLP): spaCy, BERT, NLTK, Hugging Face, NER, Sentiment Analysis, Text Classification, Conversational AI, Document Parsing, OCR (Tesseract, AWS Textract)

MLOps & Model Deployment: MLflow, Docker, Kubernetes, TorchServe, Triton Inference Server, ONNX, TorchScript, Model Quantization (INT8, 4-bit), CI/CD (GitHub Actions, Jenkins), Canary & Shadow Deployment

DevOps & Workflow Automation: Docker, Kubernetes, GitHub Actions, Jenkins, Apache Airflow, DVC, CI/CD Automation

Big Data & Databases: Presto, Hive, Scuba, Spark SQL, Apache Spark, PySpark, ETL Pipelines

Distributed Systems: Petabyte-scale Data Logs, DeepSpeed, NCCL, Multi-GPU Training

Cloud & Infrastructure: AWS (Lambda, S3, SageMaker, CloudWatch), Google Cloud (Vertex AI), Azure ML Studio, Meta MTIA Chips, NVIDIA Grace Hopper Superchip

Responsible AI & Governance: SHAP, LIME, Fairlearn, AI Watermarking, Brand Safety Filters, Bias Detection, Human-in-the-loop, GDPR/RBI/KYC/AML

Visualization & Analytics: Power BI, Tableau, Matplotlib, Seaborn, Business Intelligence Dashboards

Frameworks & APIs: Flask, FastAPI, RESTful APIs, ERP Integration (SAP, Oracle, NetSuite)

Software Engineering & Practices: Agile/Scrum, A/B Testing, Software Architecture, Git (GitHub, GitLab, Bitbucket), Code Reviews, System Integration

Development Environments & Tools: VSCode, Jupyter Notebook, Git, GitLab, Bitbucket, Linux, Containerization Environments

EDUCATION

Masters Of Science in Data Analytics
Northeastern University

MSc In Biotechnology
Deogiri College Aurangabad, MH, India

Bsc In Agri Biotechnology
Shri Shivaji College of Agri Biotechnology, Amravati, MH, India

PROJECT

Enterprise RAG Pipeline with Snowflake Cortex AI

- Designed and deployed a **retrieval-augmented generation (RAG) pipeline** using **Snowflake Cortex AI embeddings + vector search** to improve enterprise chatbot and document summarization accuracy.
- Fine-tuned **LLaMA-2/3 models** on enterprise knowledge bases with Hugging Face Transformers, reducing hallucination rates by **27%**.
- Built hybrid search with **BM25 + dense embeddings**, improving factual correctness in responses.
- Deployed pipeline using **Kubernetes, Docker, and MLflow**, achieving **<100ms inference latency** at scale.
- Implemented **data governance controls** with role-based access, ensuring GDPR/KYC compliance for enterprise data usage.

Real-Time Recommendation Engine with Snowpark ML

- Developed a **real-time recommendation system** leveraging **Snowpark ML** and **PyTorch**, processing **50+ TB of structured/unstructured data daily** from multi-cloud sources (AWS S3, Azure Data Lake, GCP BigQuery).
- Served personalized recommendations to **300+ enterprise users** through **FastAPI + TorchServe microservices**.
- Optimized models with **ONNX runtime and INT8 quantization**, reducing inference costs by **35%** while maintaining accuracy.
- Automated feature engineering pipelines with **Airflow + dbt**, cutting batch latency by **40%**.
- Integrated **Grafana dashboards** for real-time monitoring of engagement metrics, improving system adoption by product teams.

Responsible AI Monitoring & Governance Platform

Built a **governance and monitoring platform** for deployed ML models using **MLflow, Prometheus, and Grafana**, enabling proactive tracking of **model drift, fairness, and bias**.

- Integrated **SHAP, LIME, and Fairlearn** for explainability and fairness audits across models in production.
- Automated **retraining triggers** based on drift thresholds using **Airflow CI/CD workflows**, ensuring continuous compliance.
- Partnered with legal and compliance teams to embed **GDPR, AML, and KYC regulatory checks** into ML workflows.
- Achieved **99.99% uptime** of AI-powered services with robust monitoring and high-availability deployments.